

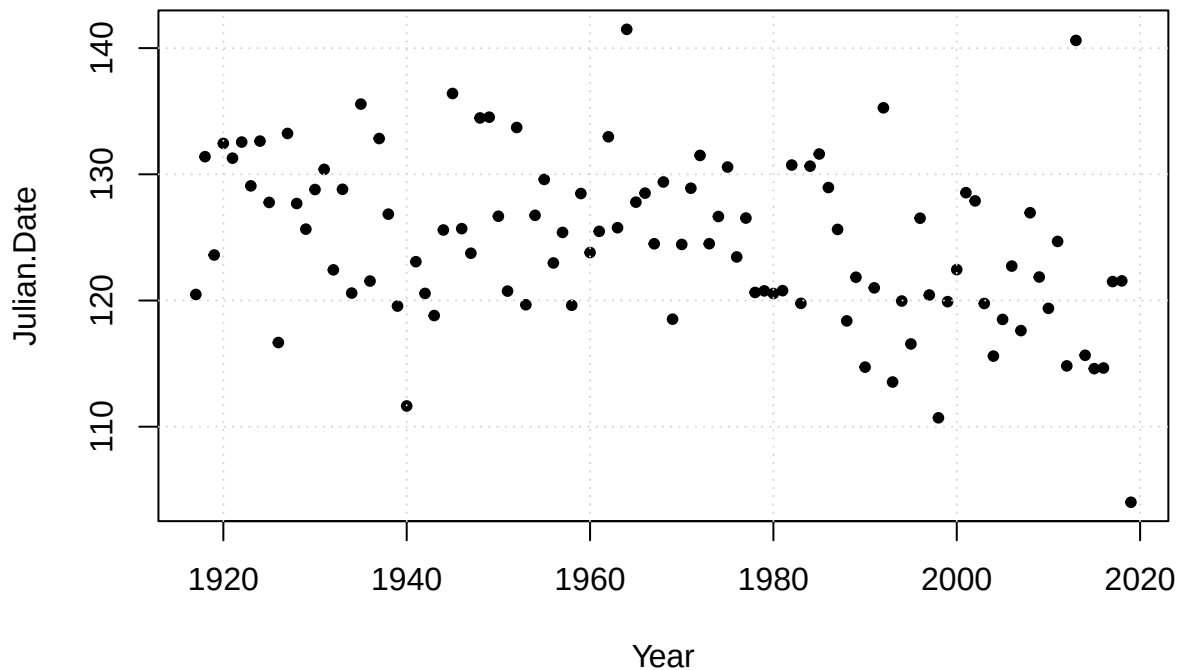
Change point analysis: Nenana

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Reading the data

```
nenana<-read.csv("nenana.csv")  
plot(Julian.Date~Year,data=nenana,pch=20)  
grid()
```

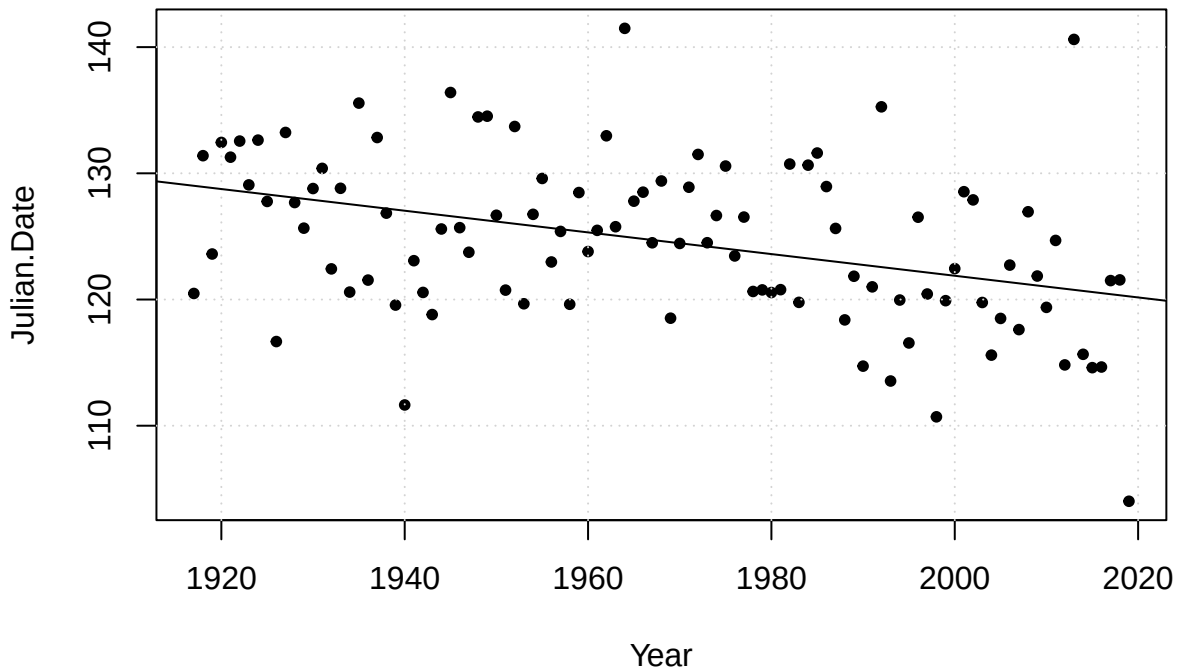


Simple linear model

```
fit<-lm(Julian.Date~Year,data=nenana)  
summary(fit)
```

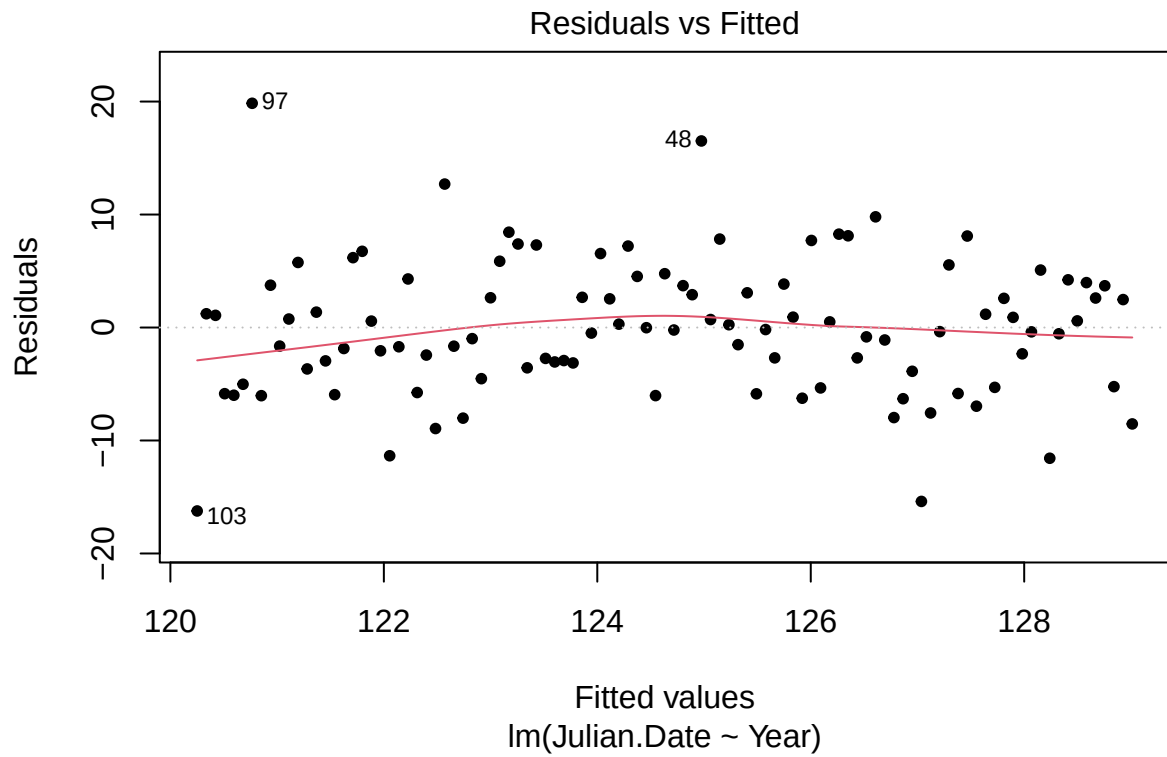
```
##  
## Call:
```

```
## lm(formula = Julian.Date ~ Year, data = nenana)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.2365  -3.7701  -0.1806   3.7946  19.8454
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 293.6811    39.5653   7.423 3.71e-11 ***
## Year        -0.0859     0.0201  -4.273 4.37e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.066 on 101 degrees of freedom
## Multiple R-squared:  0.1531, Adjusted R-squared:  0.1447
## F-statistic: 18.26 on 1 and 101 DF,  p-value: 4.372e-05
plot(Julian.Date~Year,data=nenana,pch=20)
grid()
abline(fit)
```

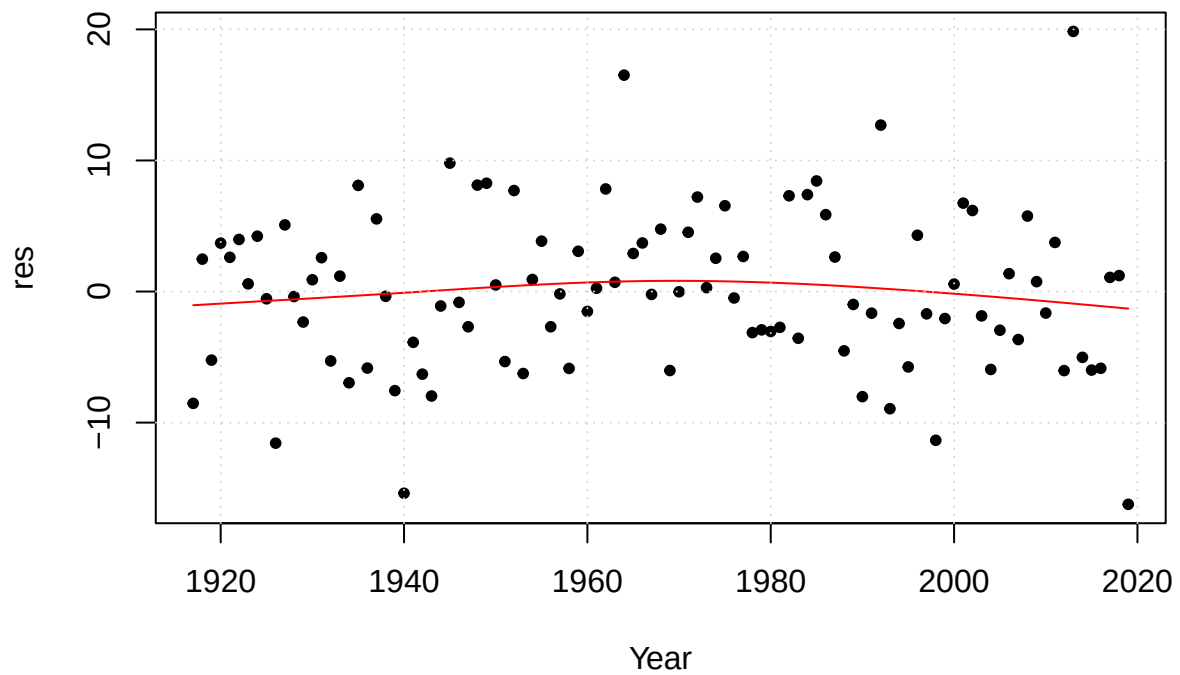


Residuals

```
plot(fit,which=1,pch=20)
```



```
res<-resid(fit)
attach(nenana)
plot(res~Year,pch=20)
a<-smooth.spline(Year,res)
lines(a,col="red")
grid()
```



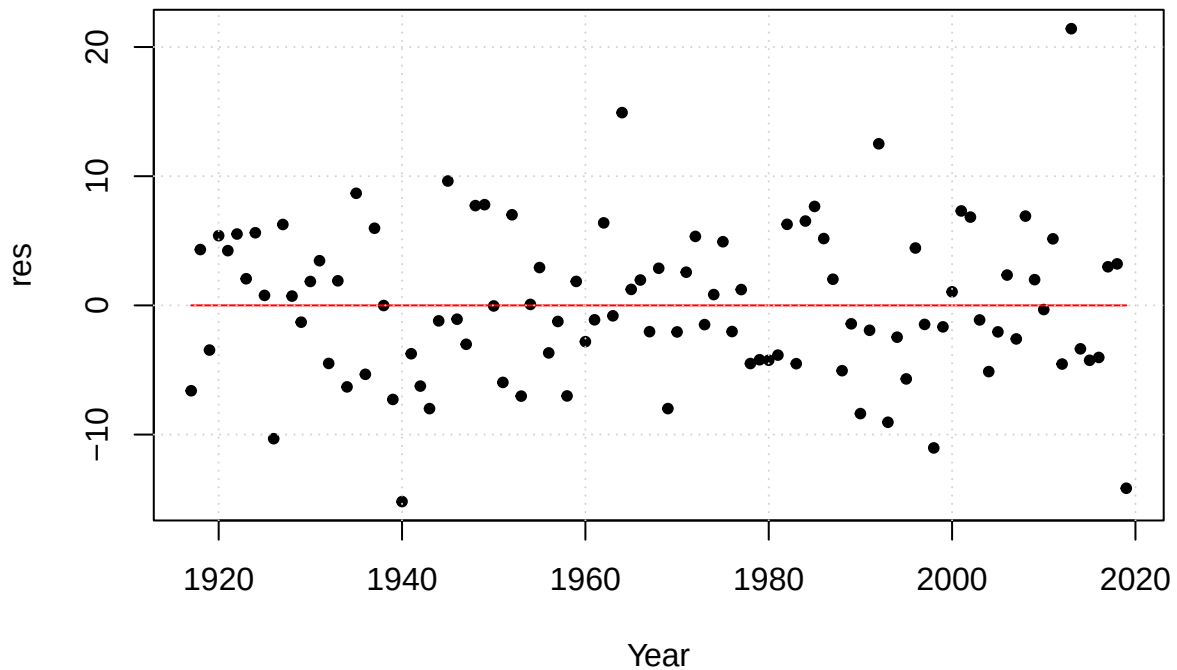
Model with changepoint

```
changepoint<-1970
nenana$cp1970<-pmax(0,nenana$Year-changepoint)
fit.c<-lm(Julian.Date~Year+cp1970,data=nenana)
summary(fit.c)

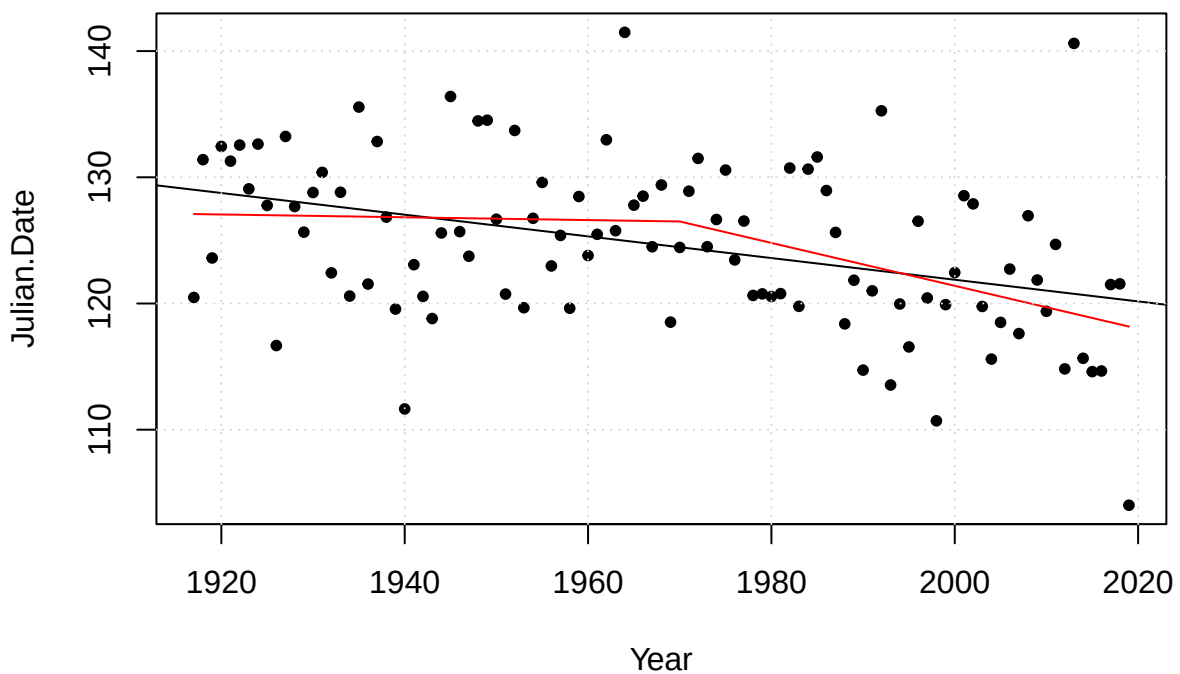
##
## Call:
## lm(formula = Julian.Date ~ Year + cp1970, data = nenana)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1878  -4.1186  -0.8086   4.2791  21.4204
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 148.39396   82.44579   1.800   0.0749 .
## Year        -0.01111    0.04232  -0.263   0.7934
## cp1970       -0.15882    0.07941  -2.000   0.0482 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.978 on 100 degrees of freedom
## Multiple R-squared:  0.1857, Adjusted R-squared:  0.1694
## F-statistic: 11.4 on 2 and 100 DF, p-value: 3.465e-05
```

Residuals

```
res<-resid(fit.c)
plot(res~Year,pch=20)
a<-smooth.spline(Year,res)
lines(a,col="red")
grid()
```



```
plot(Julian.Date~Year,data=nenana,pch=20)
abline(fit)
lines(nenana$Year,predict(fit.c),col="red")
grid()
```



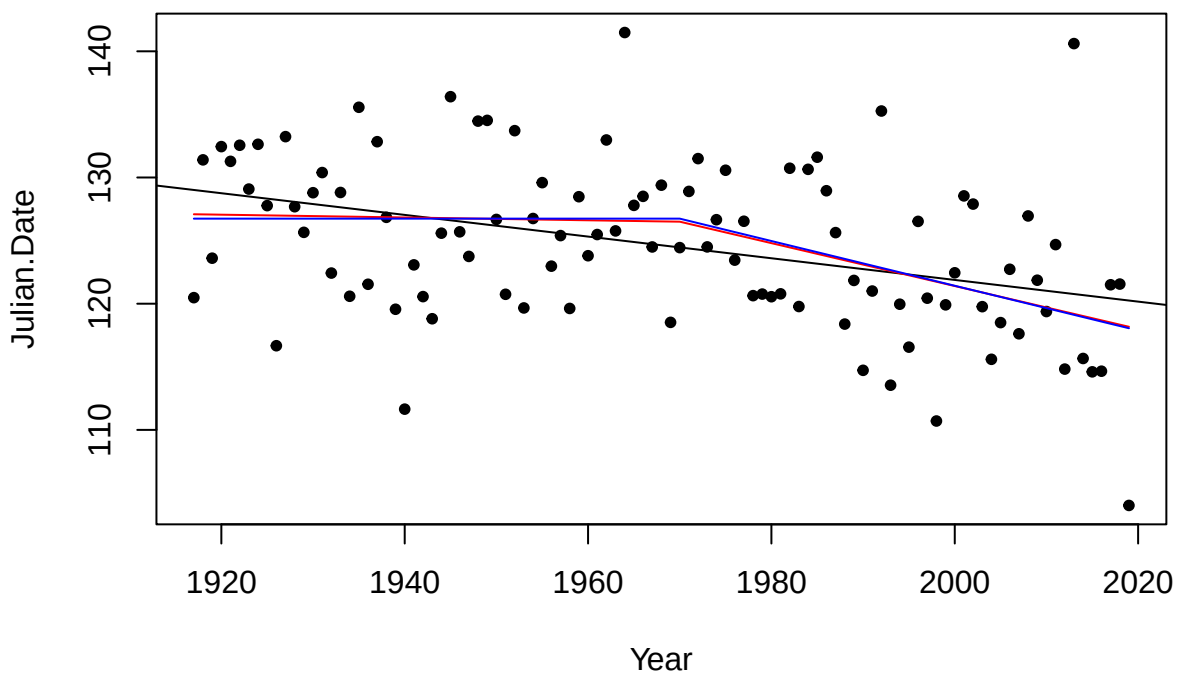
Second model with changepoint

No influence with of the time before 1970 ?

```
fit.c2<-lm(Julian.Date~cp1970,data=nenana)
summary(fit.c2)
```

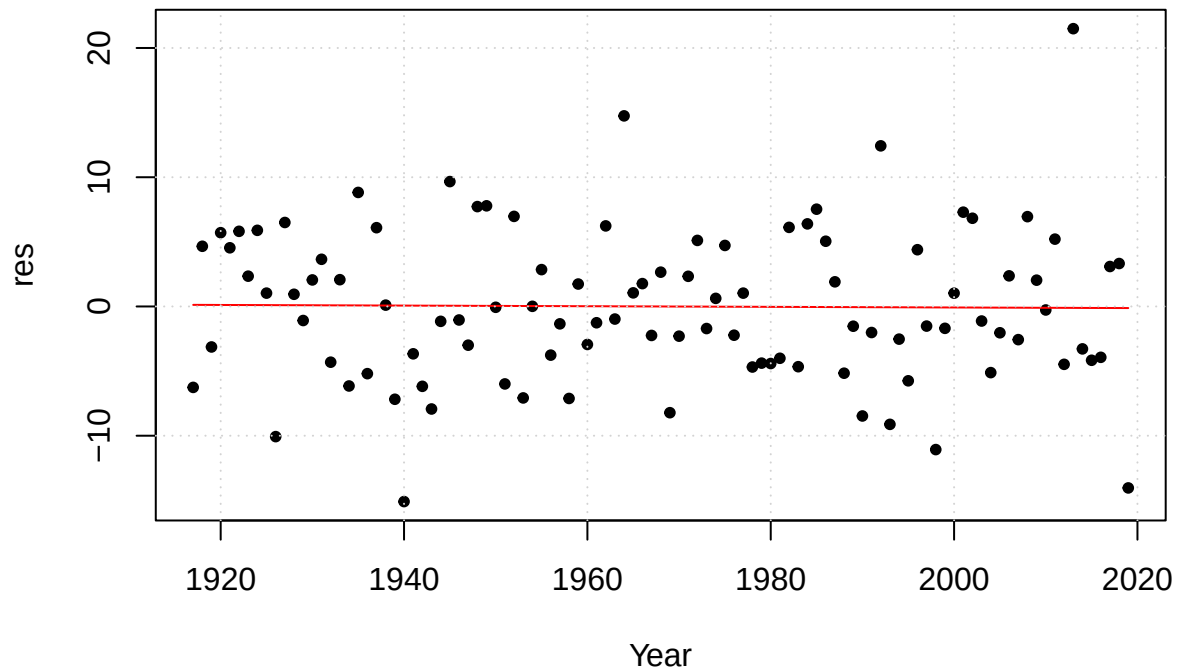
```
##
## Call:
## lm(formula = Julian.Date ~ cp1970, data = nenana)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.0958  -4.0865  -0.9722   4.4654  21.4935
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 126.7399     0.7331  172.89 < 2e-16 ***
## cp1970      -0.1772     0.0370   -4.79 5.73e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.95 on 101 degrees of freedom
## Multiple R-squared:  0.1851, Adjusted R-squared:  0.1771
## F-statistic: 22.94 on 1 and 101 DF, p-value: 5.731e-06
```

```
plot(Julian.Date~Year,data=nenana,pch=20)
abline(fit)
lines(nenana$Year,predict(fit.c),col="red")
lines(nenana$Year,predict(fit.c2),col="blue")
```



Residual analysis

```
res<-resid(fit.c2)
plot(res~Year,pch=20)
a<-smooth.spline(Year,res)
lines(a,col="red")
grid()
```



```
qqnorm(res)  
qqline(res,col="red")
```

Normal Q-Q Plot

